

Introduction

Computing the low-lying spectrum of quantum Hamiltonians is a central challenge in many-body physics and remains an active area of research across fields ranging from condensed matter to nuclear and particle physics. We develop a continuous-time Monte Carlo method for computing the low-lying spectrum of generic quantum Hamiltonians defined on finite Hilbert spaces. Unlike discrete-time formulations, the approach we develop avoids Trotter discretization errors and provides direct access to the low-energy subspace. While the method is reliable in systems that do not suffer from sign problems, it can still yield useful information about the low-energy spectrum even in the presence of sign problems on small system sizes. To demonstrate these features, we apply the method to the Shastry-Sutherland model on small lattices and benchmark our results against exact diagonalization. Our results show that the method can serve as a valuable tool for exploring strongly correlated systems where exact solutions are inaccessible.

Geometric Frustration and The Shastry-Sutherland Model

- The Shastry-Sutherland Model (SSM) is given by an anisotropic spin-1/2 Heisenberg Hamiltonian on a lattice:

$$H_{SSM} = J' \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J \sum_{\langle\langle i,j \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j,$$

with nearest neighbor and next nearest neighbor interactions. J' and J are positive. Initially, it was a "Toy Model" for frustration in a 2D-lattice.

- The ground state of H_{SSM} is exactly known to be the product of *orthogonal dimer singlets*

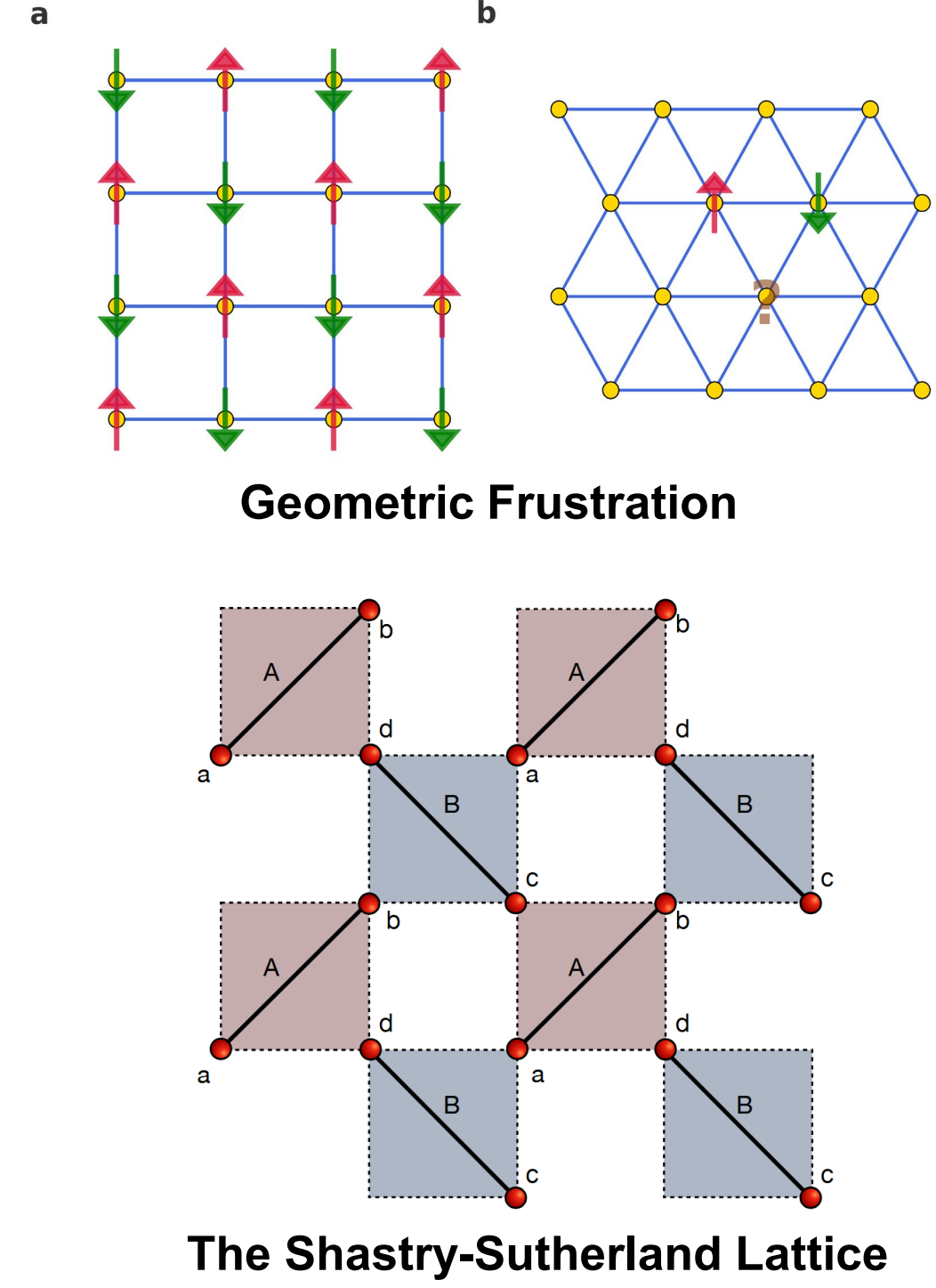
$$|\psi_{dimer}\rangle = \prod |S_{\langle i,j \rangle}\rangle$$

with $|S\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ for $J'/J \leq 0.5$.

- In 1999, H_{SSM} was discovered to be applicable in describing the physics of $\text{SrCu}_2(\text{BO}_3)_2$. Since then, series of materials have been confirmed to be modeled by the same model.

- Experimental evidence of a possible intermediate phase around $J'/J \approx 0.7$ between dimerized and Neel ordered state. The nature and existence of this phase continues to be debated.

- Frustration makes the lattice *non-bipartite* due to odd cycles. Hence, the Hamiltonian cannot be made *stoquastic* (all weights non-negative) and the Monte Carlo suffers from the notorious "sign problem".



Continuous-time Path Integral Configurations

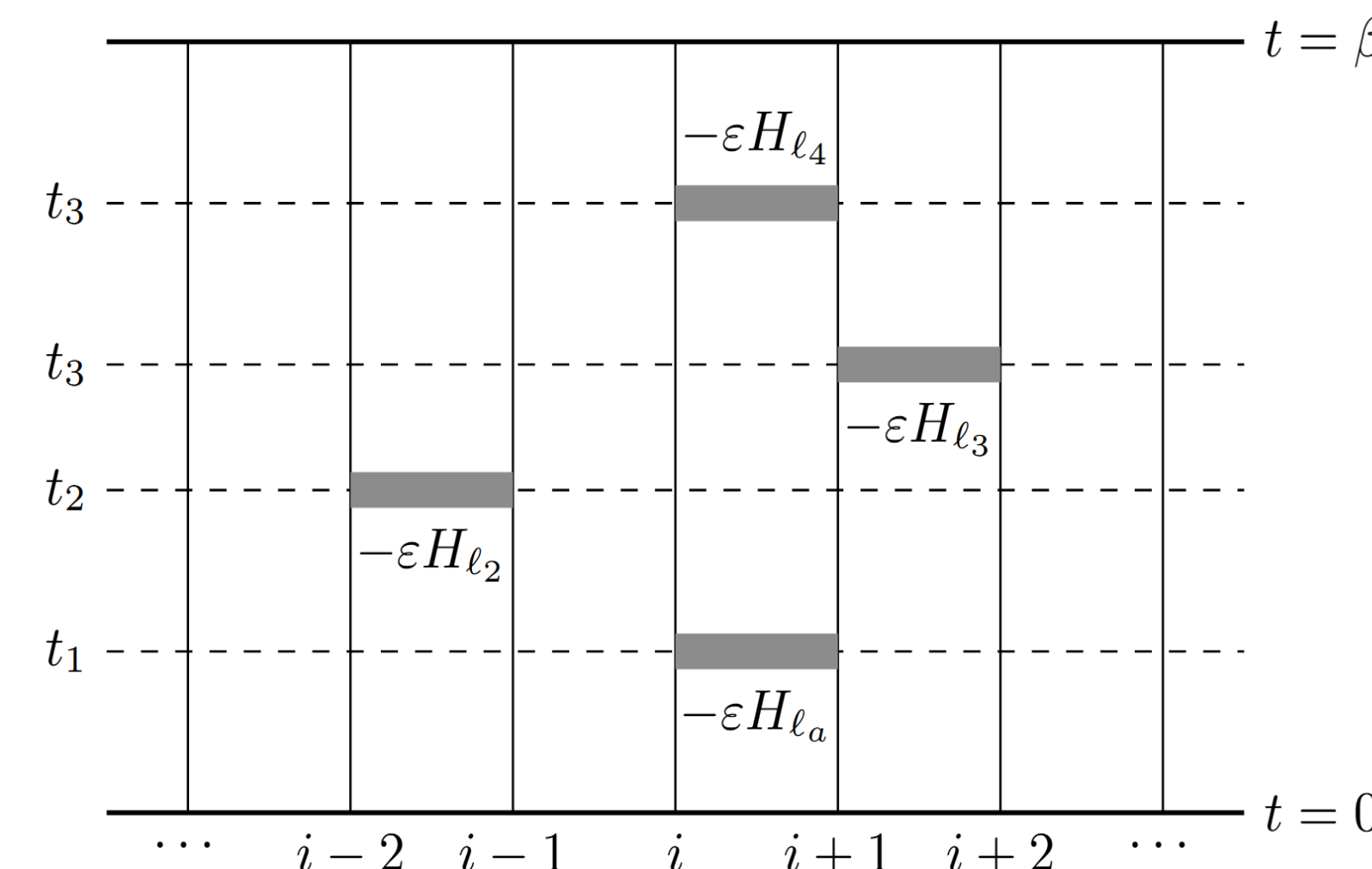


Figure: Illustration of a continuous-time path-integral configuration [C].

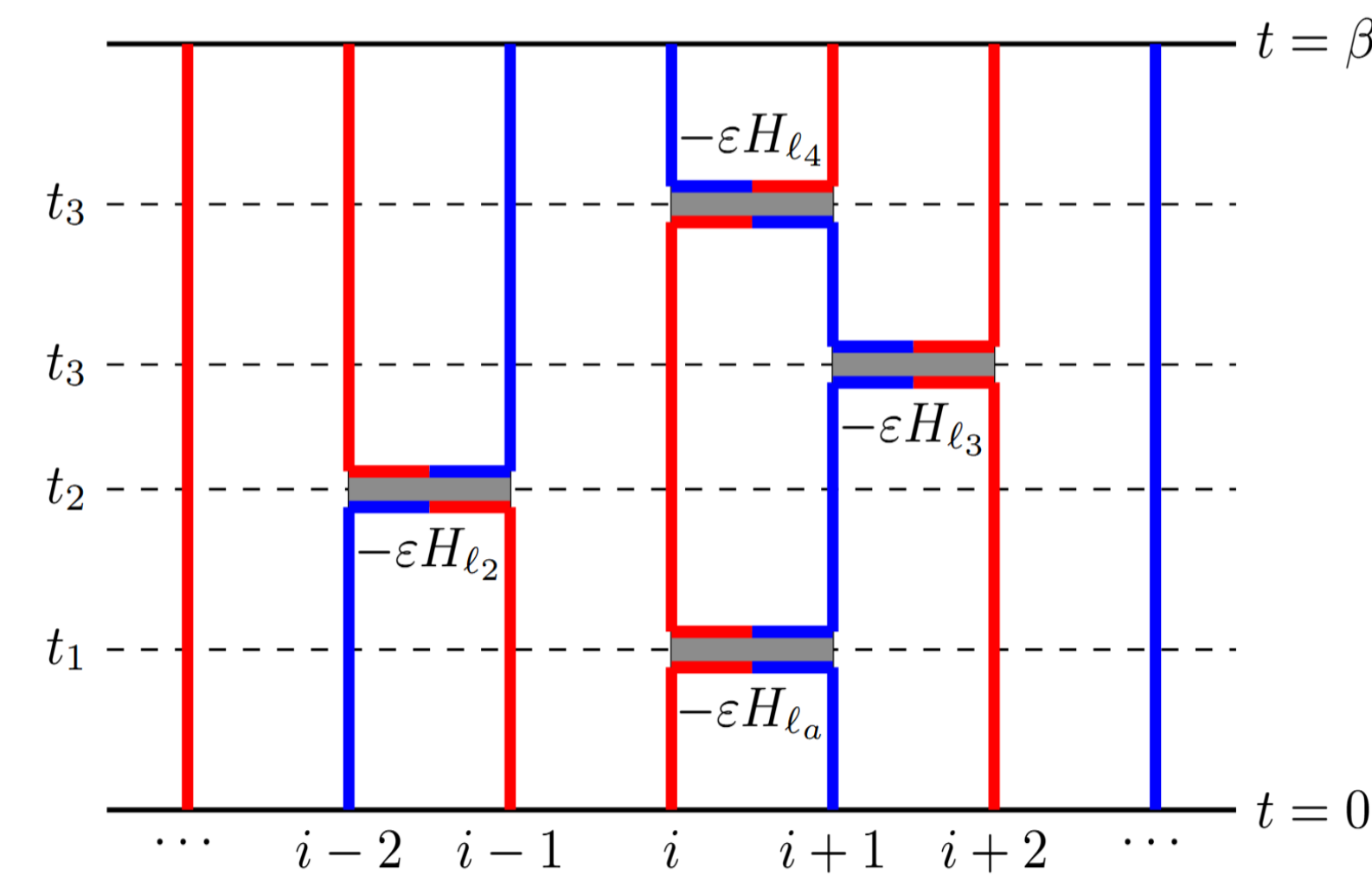


Figure: The loop-cluster configuration associated with the configuration [C].

Choosing the Reference Basis States

- The eigenvalues of the $k \times k$ matrix EZ^{-1} are guaranteed to converge to the correct low energy eigenvalues of H at large values of β .
- The rate of convergence depends greatly on the choice of the k basis states.
- If we can find a nice set of reference states with large overlap with specific set of low energy states and small overlap with higher energy states, then our eigenvalues quickly converge.
- However, finding such an optimal set of basis states remains significantly difficult. In general, the matrix elements become tiny as β grows and we encounter the well-known "sign problem". Along with Monte Carlo errors, the sign matrix Z can become singular and hence non-invertible, leading to large errors.

$$Q = \prod_i (2s_i^x),$$

- In the SSM an obvious symmetry is spin rotations. It is easy to see that $[H_{SSM}, Q] = 0$. We find it convenient to use the basis states that are eigenstates of Q .
- Four such basis states in the $Q = 1$ sector are as follows:

$$|\psi_1\rangle = \bigotimes_{i \in \text{even}} (|\uparrow\rangle + |\downarrow\rangle)_i \bigotimes_{j \in \text{odd}} (|\downarrow\rangle - |\uparrow\rangle)_j,$$

$$|\psi_2\rangle = \bigotimes_{\langle ij \rangle} |\psi_t\rangle_{\langle ij \rangle},$$

$$|\psi_3\rangle = \bigotimes_{\langle ij \rangle} |\psi_s\rangle_{\langle ij \rangle}, \quad x_i \text{ even}, x_j = x_i + 1,$$

$$|\psi_4\rangle = \bigotimes_{\langle ij \rangle} |\psi_t\rangle_{\langle ij \rangle}, \quad x_i \text{ even}, x_j = x_i + 1.$$

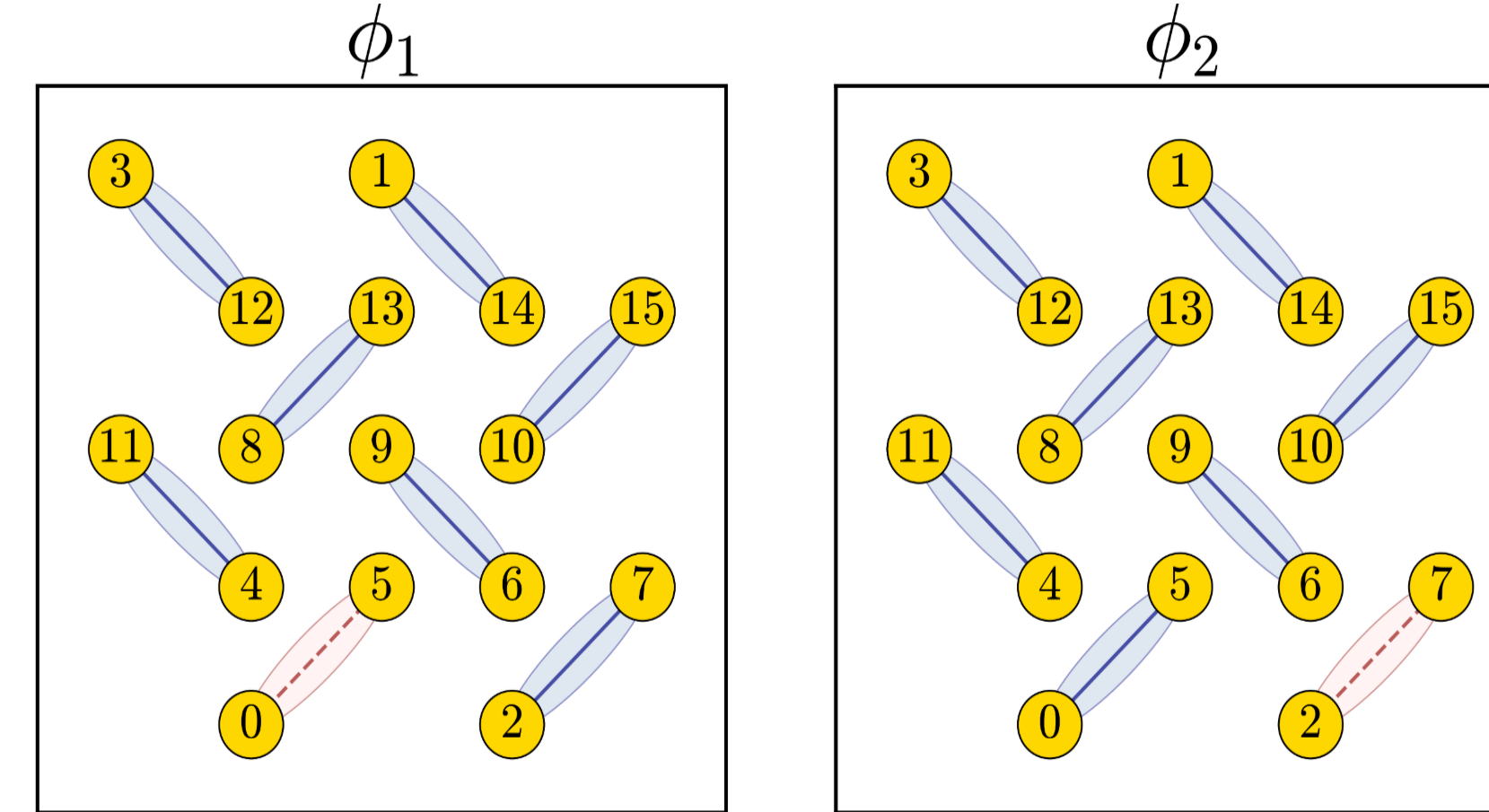
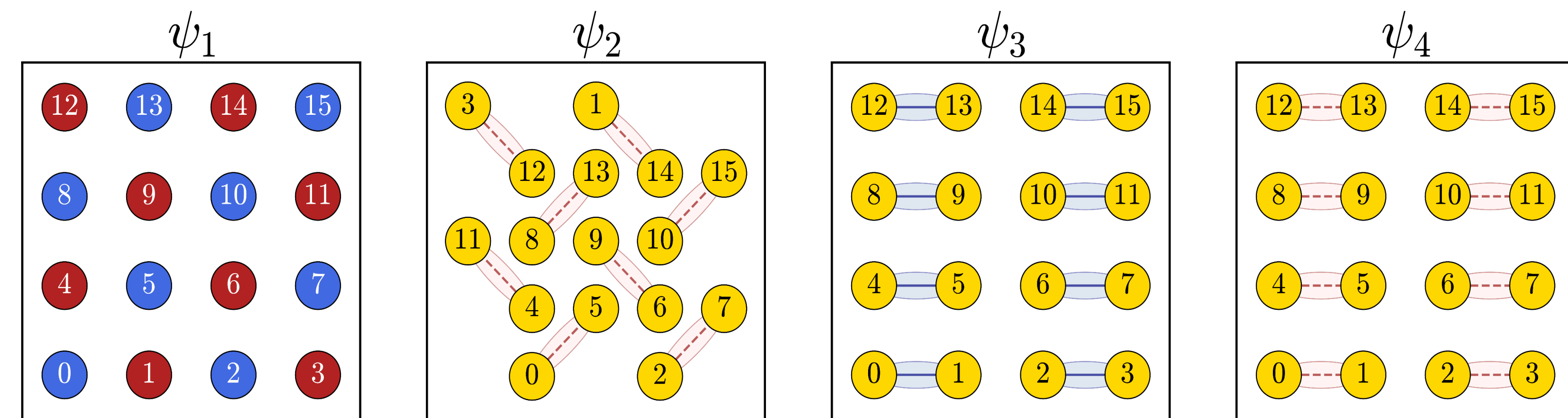


Figure: A diagrammatic representation of the $|\phi_i\rangle$ basis states on a 4×4 lattice.

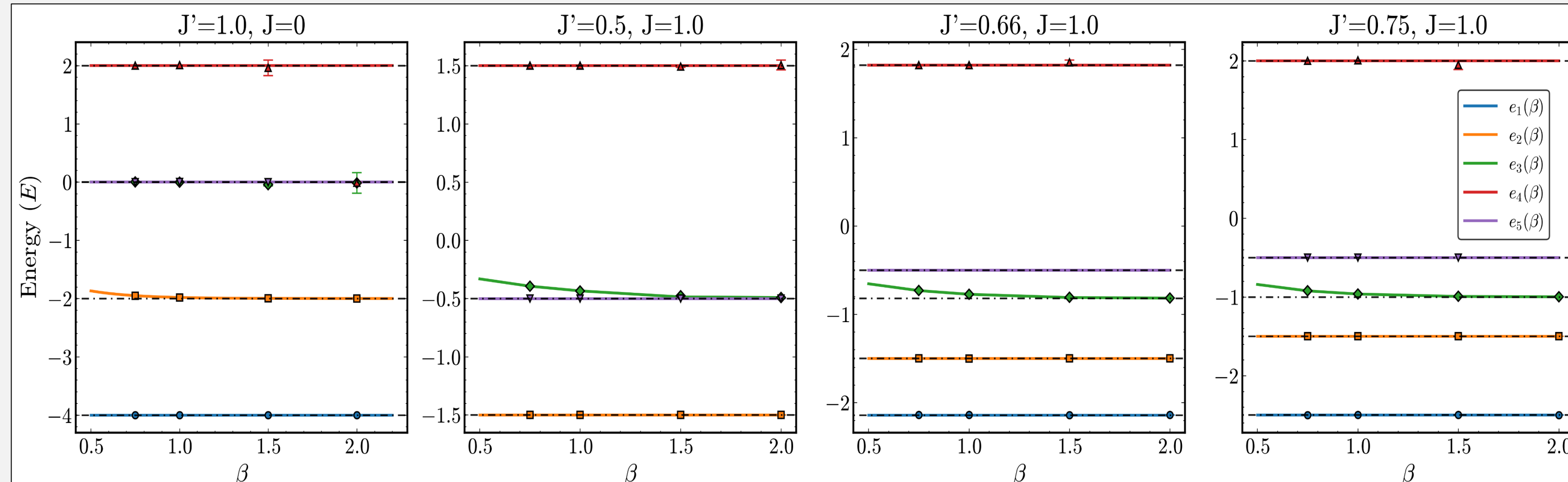
- Similarly, in the $Q = -1$ sector, we have:

$$|\phi_a\rangle = |\psi_t\rangle_{\langle ij \rangle, x_i \text{ even}} \bigotimes_{\langle ij \rangle' \neq \langle ij \rangle} |\psi_s\rangle_{\langle ij \rangle'}.$$

- At different coupling ratios J'/J , different basis states are preferred based on the overlaps with the energy eigenstates.

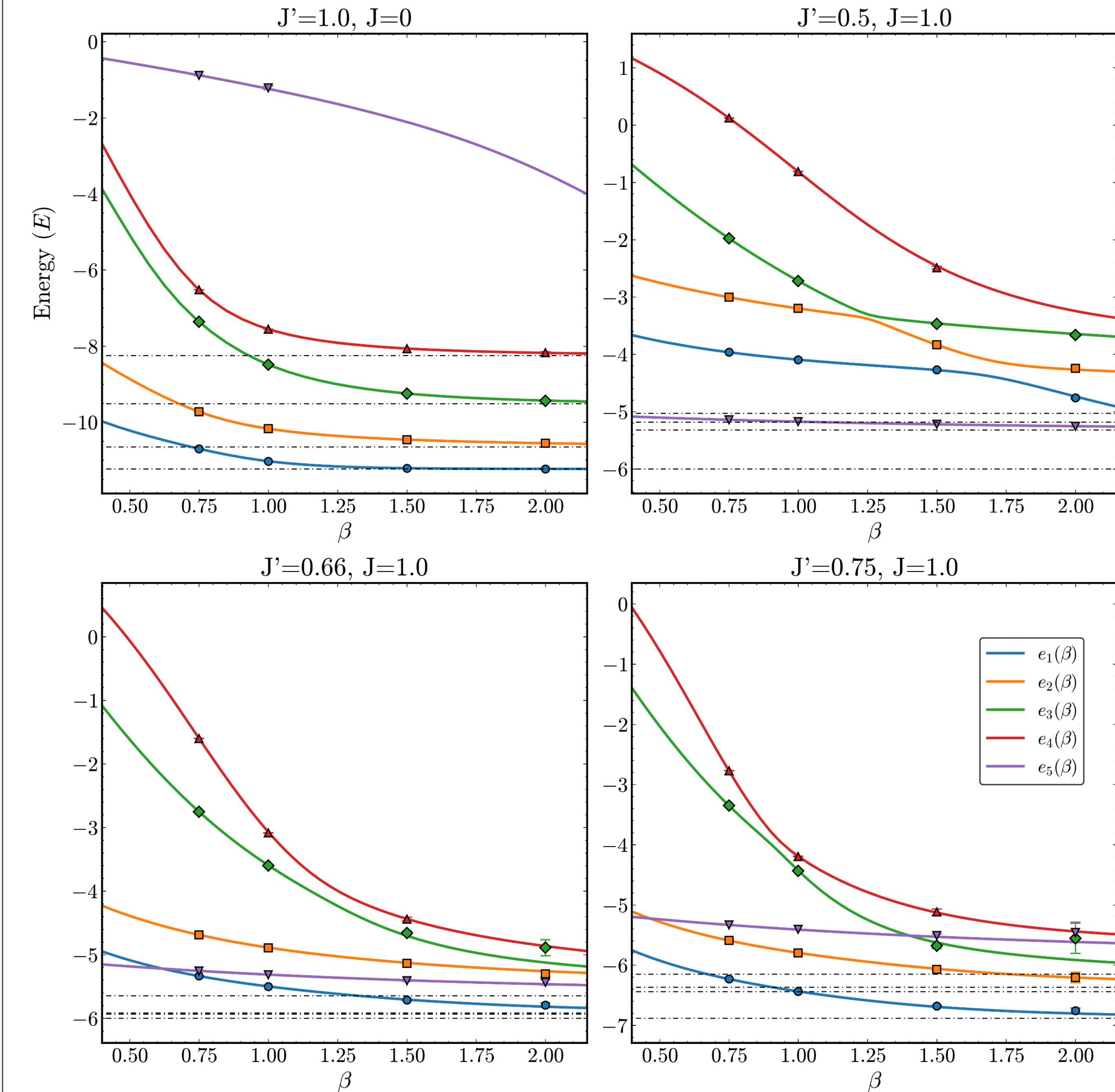


Low Energy Spectrum of the 2x2 SSM Lattice



Taken together as 5 basis states, the Monte Carlo approximates this small lattice very accurately across all coupling ratios.

Low Energy Spectrum of the 4x4 SSM Lattice



The $Q = 1$ sector basis states do well at $J' = 1, 0, J = 1$ and $J'/J = 0.66, 0.75$ and the $Q = -1$ sector basis state does well at $J'/J = 0.5$.

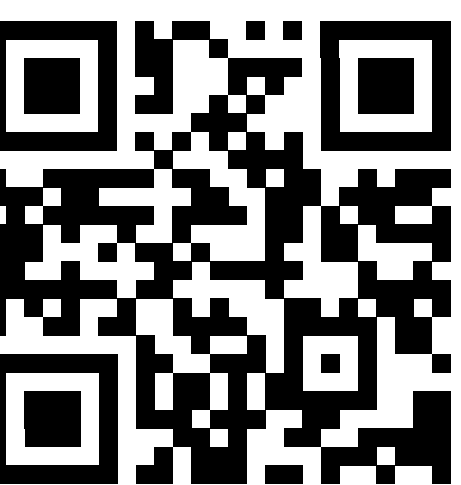
Low Energy Spectrum of the 8x8 SSM Lattice

$(J' = 1, J = 0)$				
β	$E, \{ \psi_i\rangle\}$ basis	β	$E, \{ \psi_i\rangle\}$ basis	
0.75	-41.2675(6)	2.00	-42.95(2)	
	-37.63(3)		-42.54(2)	
	-31.229(5)		-42.3(1)	
	-21.9(9)		-40.1(3)	
1.00	-41.8925(7)	3.00	-43.08(4)	
	-39.64(5)		-42.77(8)	
	-38.40(2)		-42.5(2)	
	-34.3(4)			
1.50	-42.609(8)	10.00	-43.0369(5)	
	-42.22(1)			
	-41.7(2)			
	-40.4(1)			
$\beta = 15.00 \quad E = -43.071(2)$				
$E_0 = -42.98(2), E_1 = -42.90(4), E_2 = -42.54(3)$				
<i>Phys. Rev. B 49, 5983 (1994)</i>				

J'	J	β	E
$\{ \phi_i\rangle\}$ basis $(J' = 0.50, J = 1)$			
0.50	1	0.75	-23.2(1)
0.50	1	1.00	-23.2(4)
$E_0 = -24$ (analytical ground state: $\frac{3}{8}JL^2$)			
$\{ \psi_i\rangle\}$ basis $(J' = 0.66, J = 1)$			
0.66	1	0.75	-21.2(5)
			-18.9(2)
			-9.6(4)
			-2(1)
$\{ \psi_i\rangle\}$ basis $(J' = 0.75, J = 1)$			
0.75	1	0.75	-23.9(9)
			-22.9(5)
			-12.5(8)

Our results start to disagree with Sauerwein and Oliveira (1994) in the unfrustrated regime ($J=0$) using the $Q = 1$ basis at $\beta = 15$. In the frustrated regime, our method works for small values of β . At $J/J = 0.5$, we find the gap decreases from 1 at $L=2$ to 0.68 at $L=4$. Further analysis for $L=8$ at $\beta = 1$, using the $Q = -1$ basis, suggests that the gap would be less than 0.68. We also find that $\langle \phi_a | e^{-\beta H} | \phi_b \rangle$ is almost proportional to δ_{ab} . And, our results show that the energies are coming down in the $Q=1$ channel and crossing the exact energy level of $E = -24$ somewhere between $J'/J = 0.66$ and 0.75 . For the next steps, we are exploring new choices of basis to formulate the Monte Carlo method such that the sign problem can be reduced.

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- B. Blossier, M. Della Morte, G. von Hippel, T. Mendes, and R. Sommer, On the generalized eigenvalue method for energies and matrix elements in lattice field theory, JHEP 04, 094, arXiv:0902.1265 [hep-lat].
- S. Chandrasekharan, S. T. Nguyen, and T. R. Richard-son, Worldline Monte Carlo method for few-body nuclear physics, Phys. Rev. C 110, 024002 (2024), arXiv:2402.15377 [nucl-th].
- R. A. Sauerwein and M. J. de Oliveira, "Ground-state properties of the spin-1/2 antiferromagnetic Heisenberg model on a square lattice by a Monte Carlo method, Phys. Rev. B 49, 5983 (1994).



Exact Diagonalization and Quantum Monte Carlo

